BRO Academy

# Internet and browser

HTML course: Lesson 1

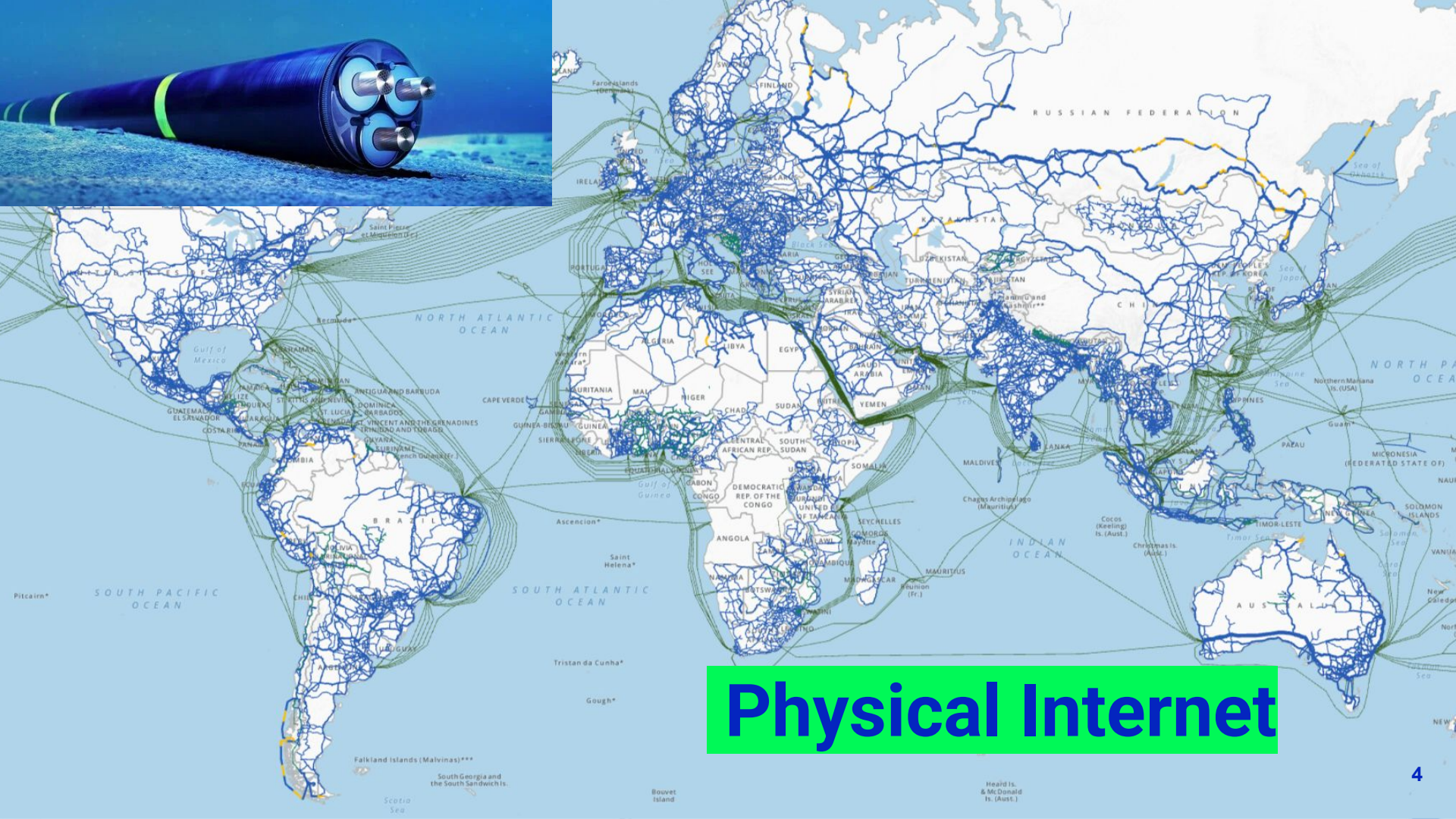
# Lesson Plan

1
How the Internet works

2
What happens when you type an URL in the browser address bar?

3
How a website is rendered in a browser

# Browser


RUSSIAN FEDERATION
NORTH ATLANTIC OCEAN
SOUTH ATLANTIC OCEAN
SOUTH PACIFIC OCEAN
INDIAN OCEAN
Physical Internet

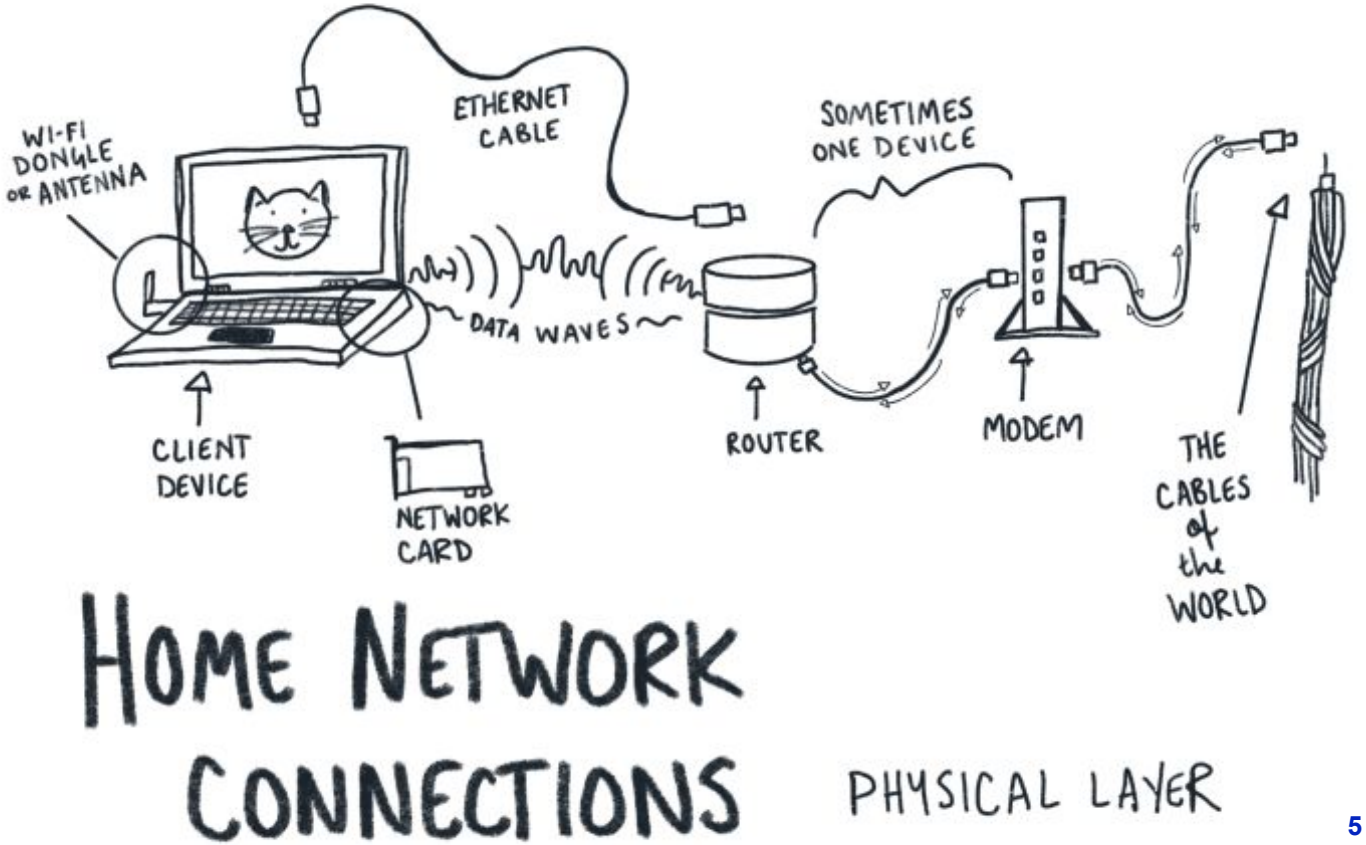
# Beginning the Journey

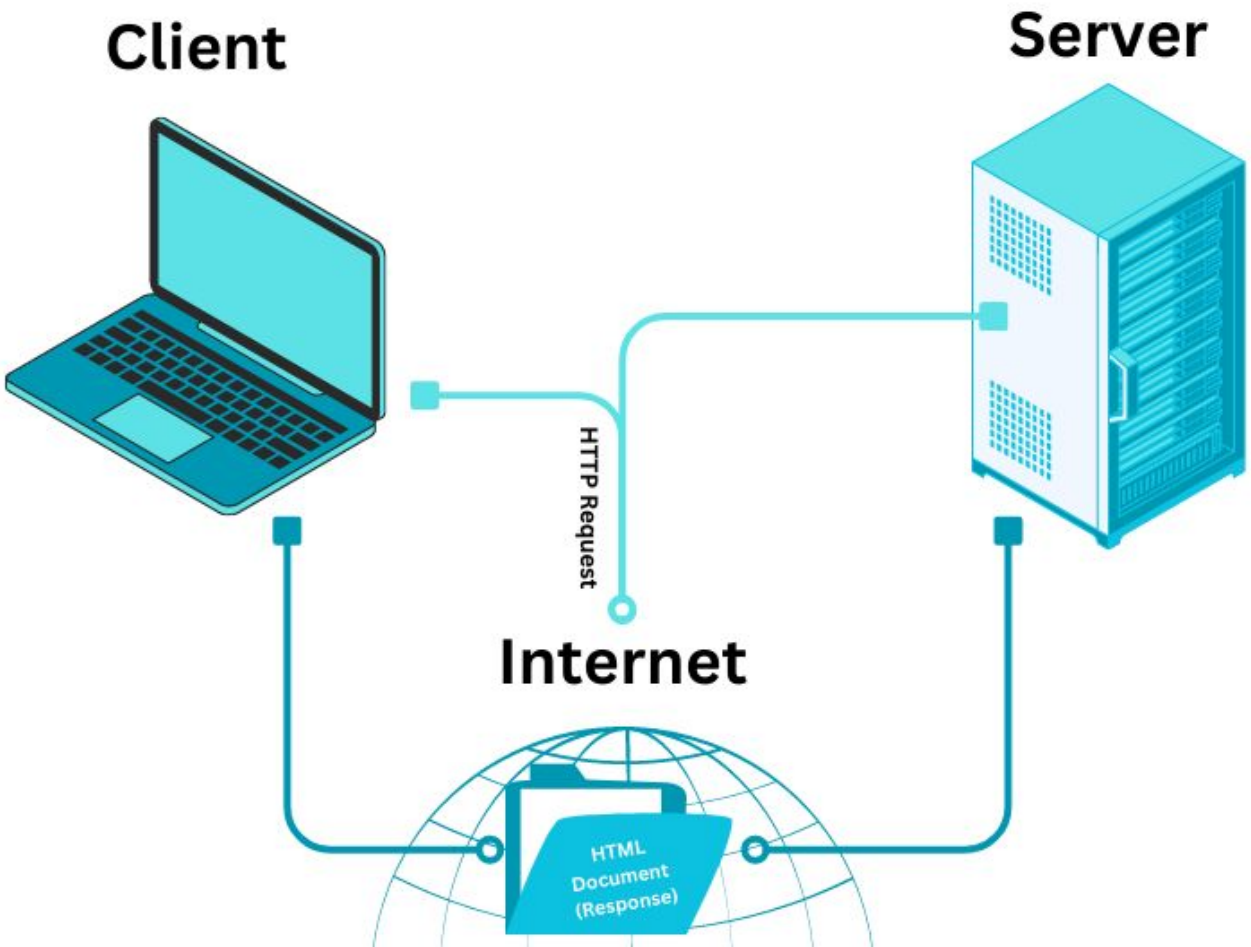
**01.** Computer

**02.** Router

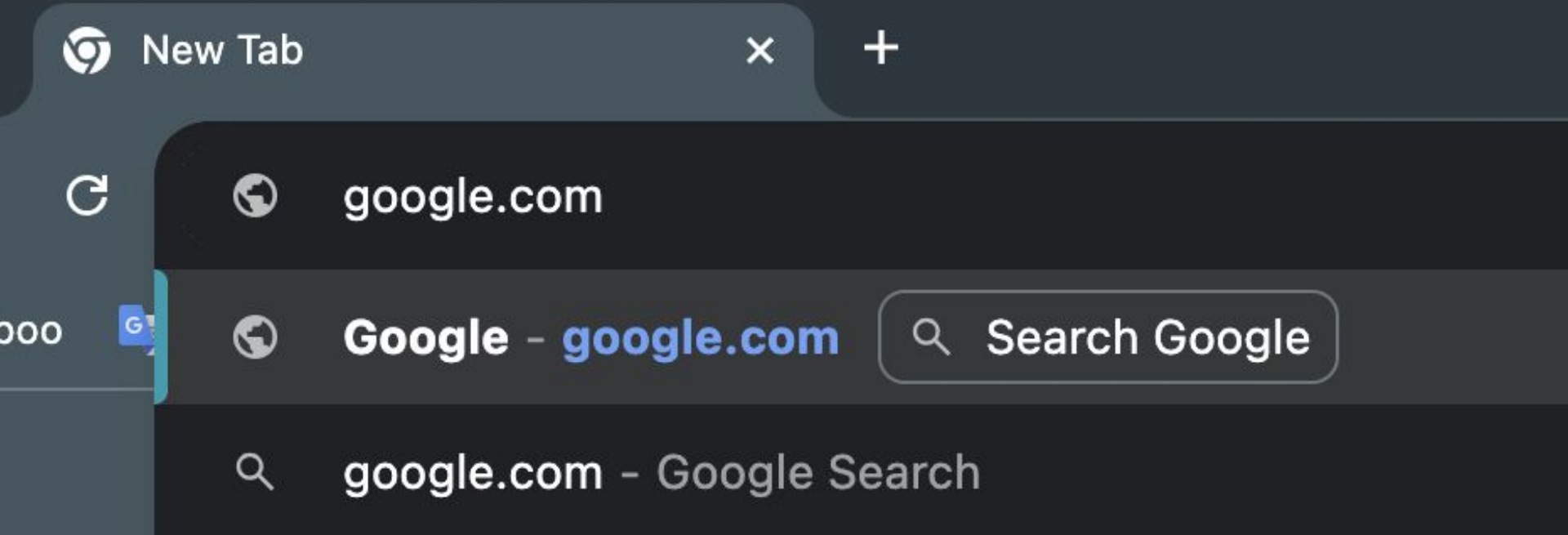
**03.** Modem

**04.** Cables




Client
Server
HTTP Request
Internet
HTML
Document
(Response)

# What happens when you type an URL in the browser address bar?



google.com = 172.253.0.0

# IP –

# Internet Protocol

your device's home address

# DNS (Domain Name System)

*Uniform resource locator*

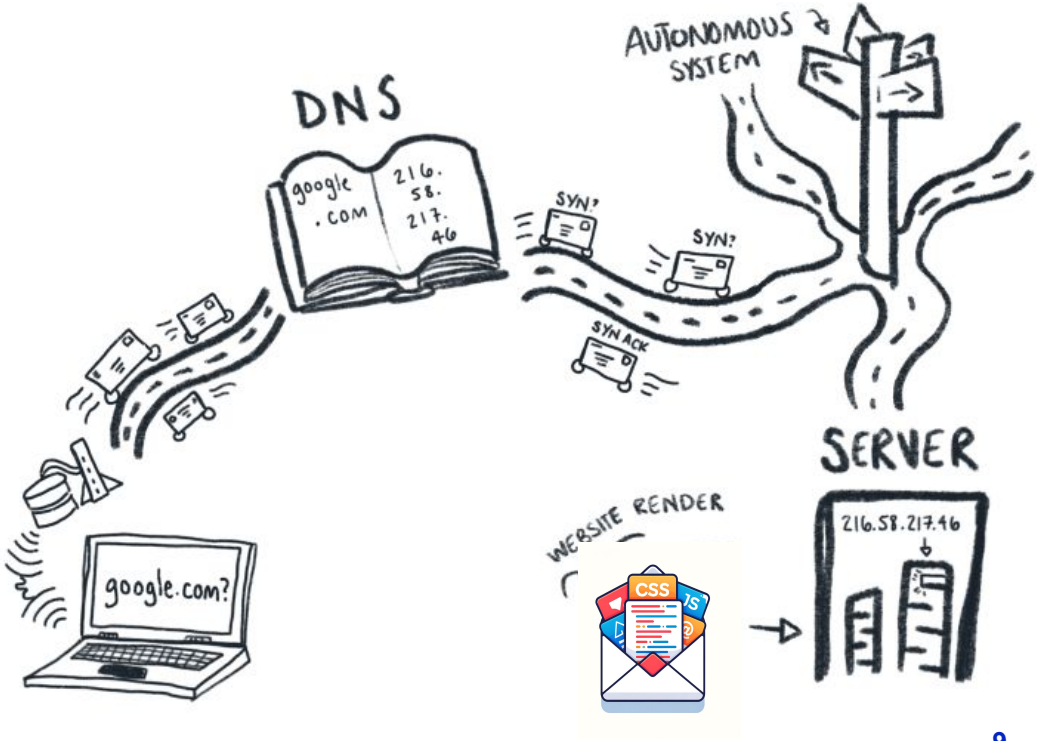
**01.** Enter **URL** in the address bar

**02.** Request from **DNS server** for IP addresses of the domain zone

**03.** Request from the next **DNS server** for **the server location** of the website

**04.** Connect to **the server** and request all **site data**

**05.** Server sends **files** to the browser



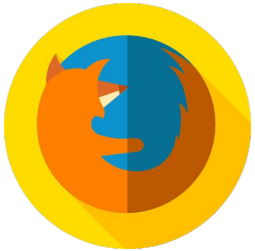
# Browsers statistics

# Browsers

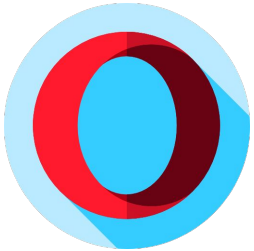
Chrome

Safari

Edge



Firefox



Opera

# Worldwide usage statistics

**Chrome**

**68%**

**Safari**
**16%**

**Edge**
5%

**Firefox**
2.6%

**Opera**
2%

**Others**
6.4%

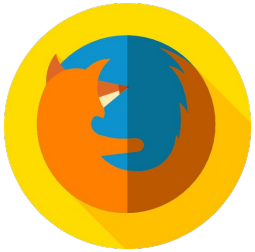
© StatCounter

Blink

Blink

Blink

Webkit



Gecko

# Browser Engines

Blink

Blink

Blink

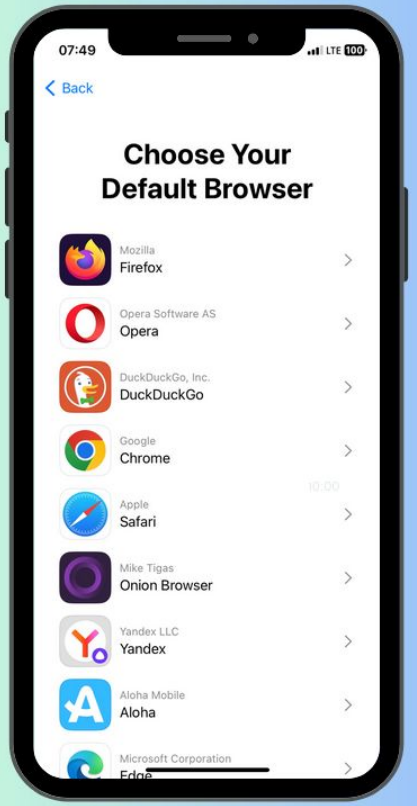
Webkit

Gecko

# Why it’s important

Browsers can render pages differently

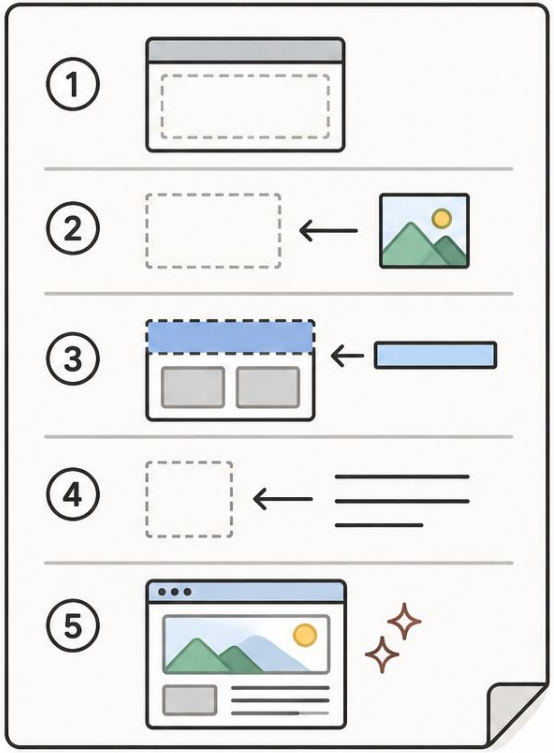
- Operational system
- Screen size
- Browser engine

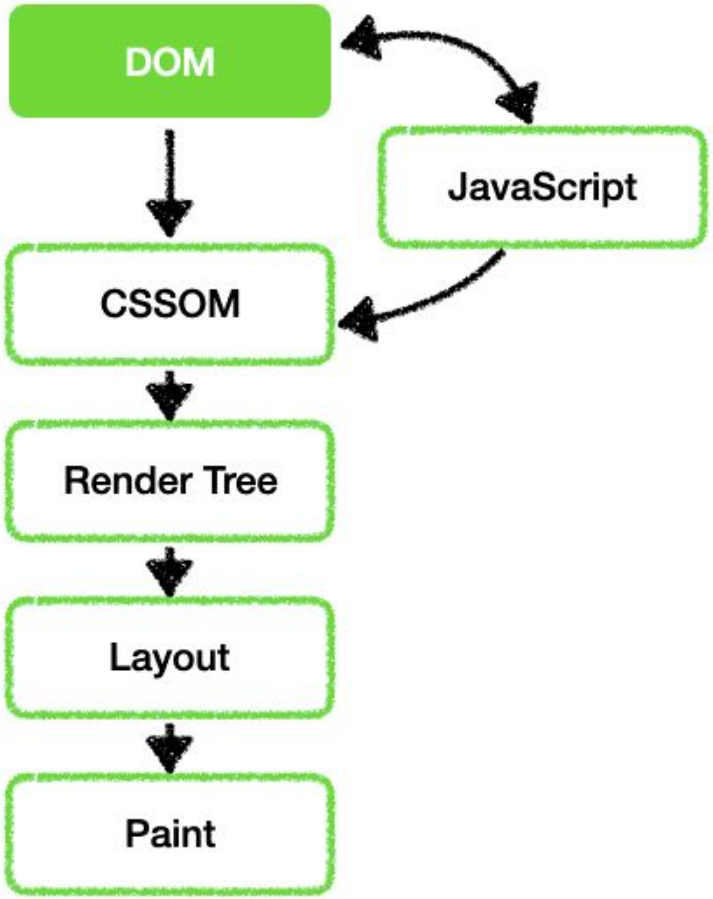


# Break time

# How a Browser Renders a website

# How a Browser Renders a Website






## How a Browser Renders a Website

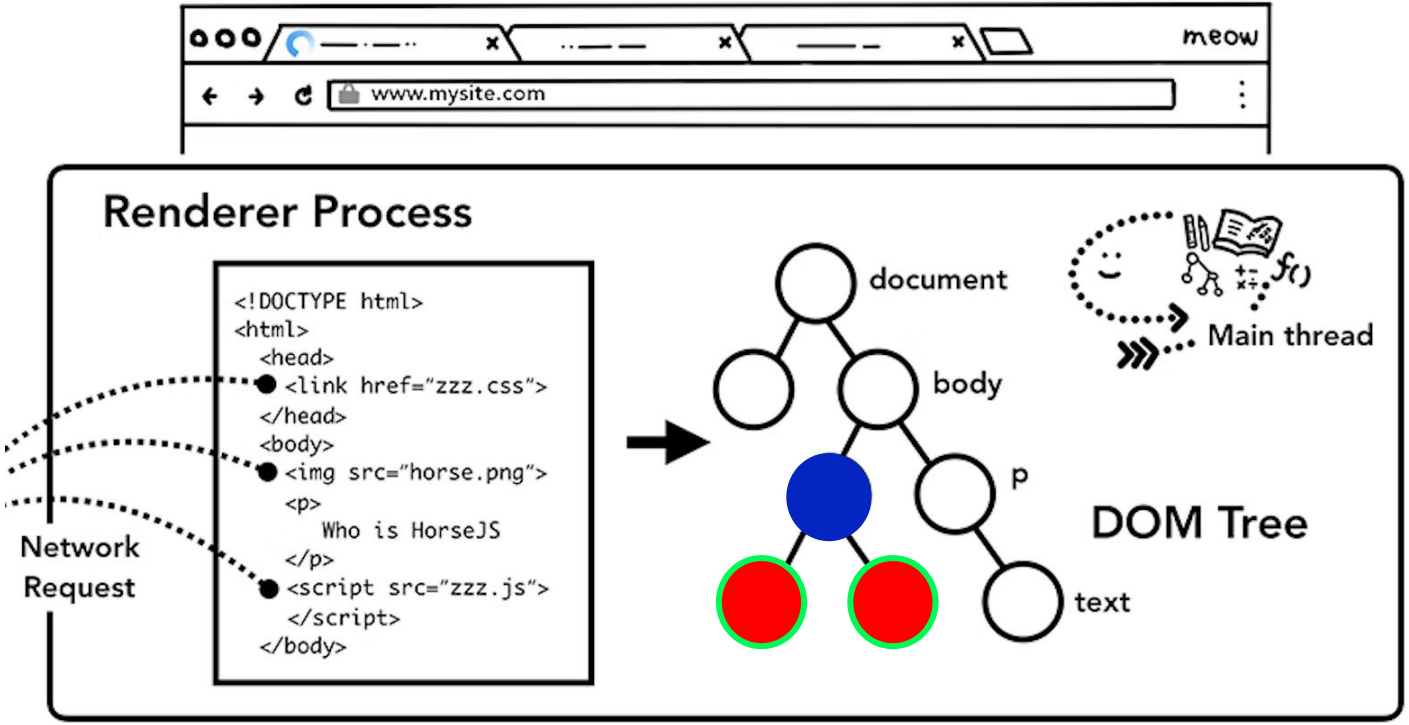
# Critical rendering path

HTML to Node Tree

# DOM

Document
Object
Model

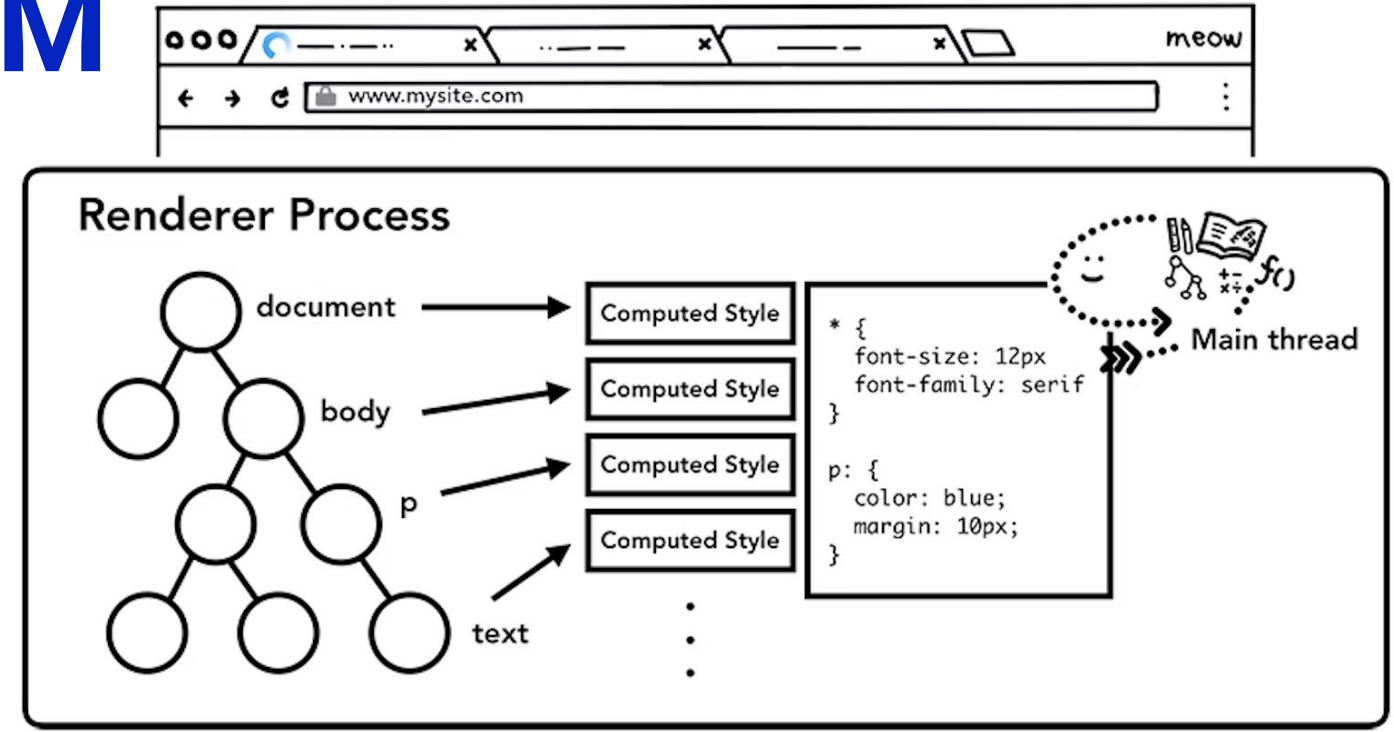
- parent
- child
- sibling



Define styles for tree elements

# CSSOM

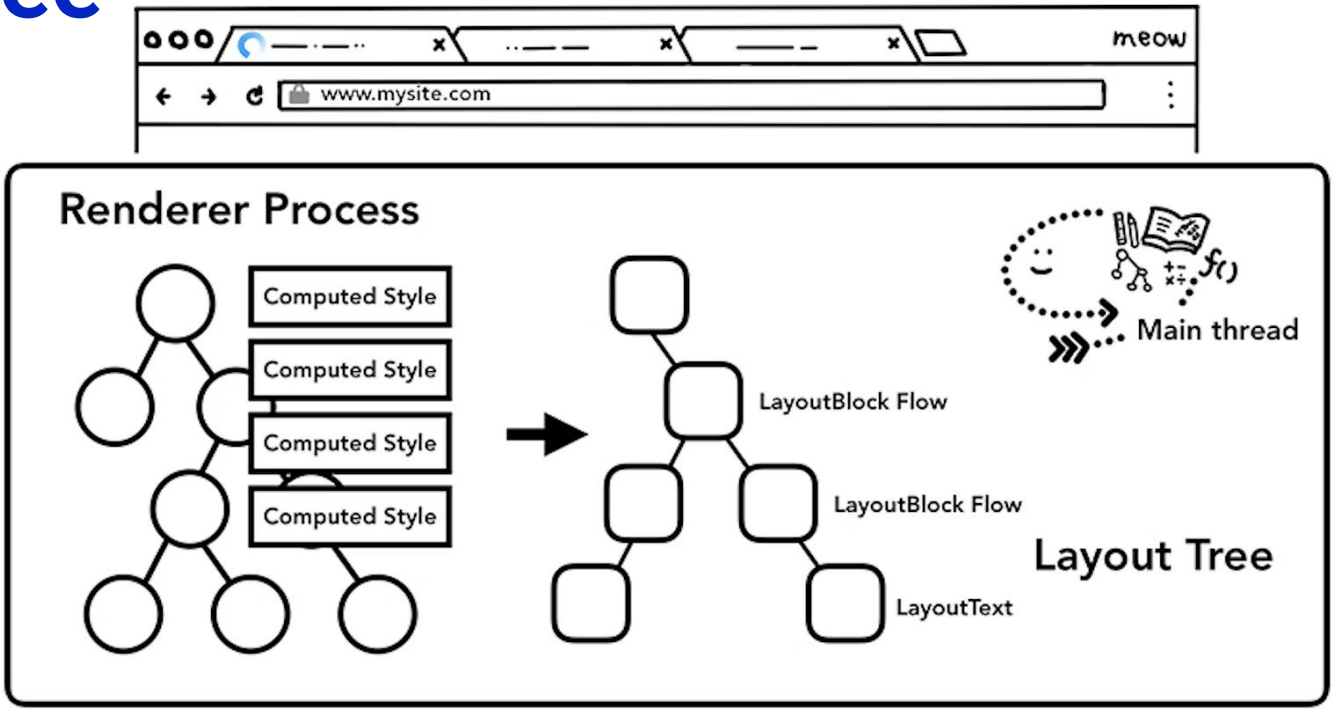
**CSS Object Model**



Combine DOM and CSSOM

# Render tree

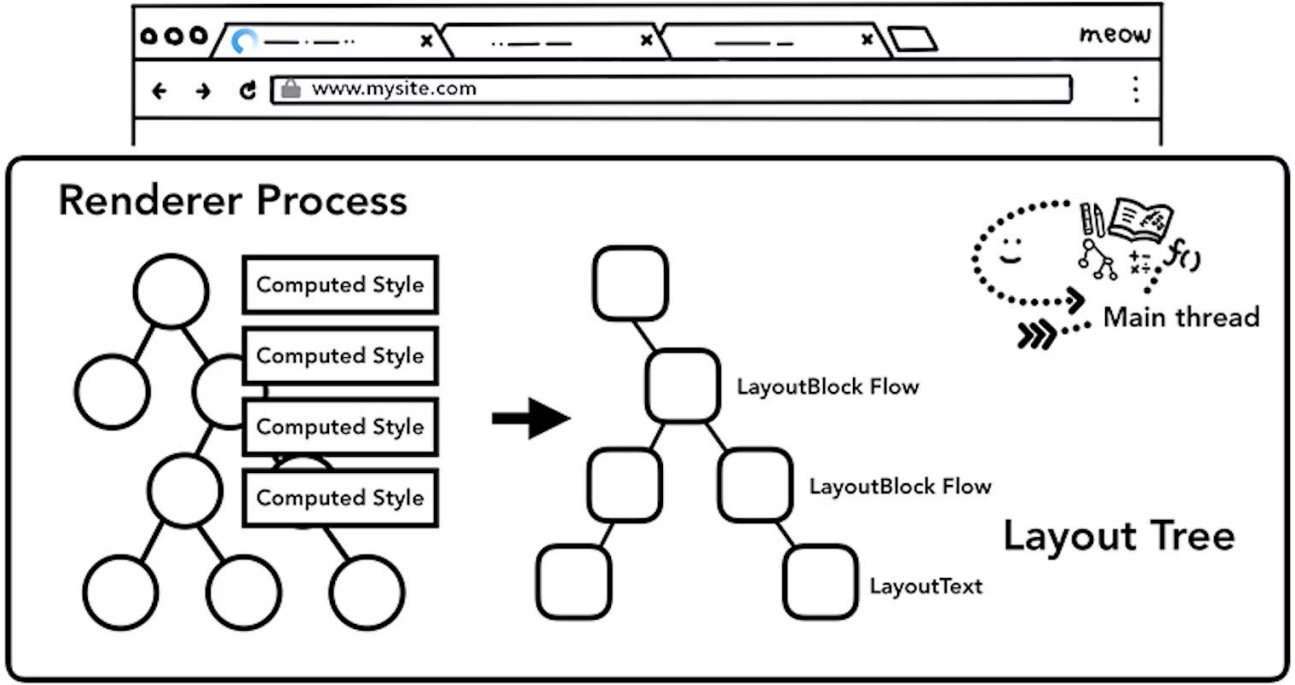
Map HTML elements to CSS styles



# Layout

Calculate space for elements

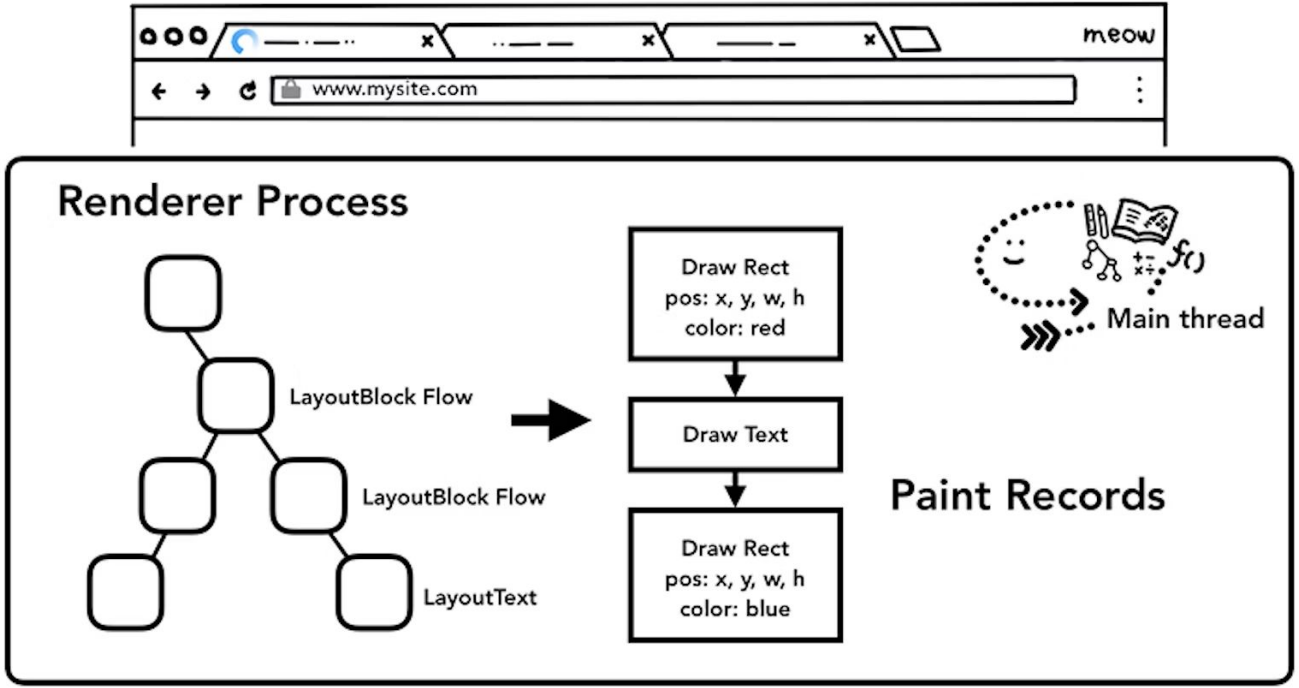
Only considers visible elements

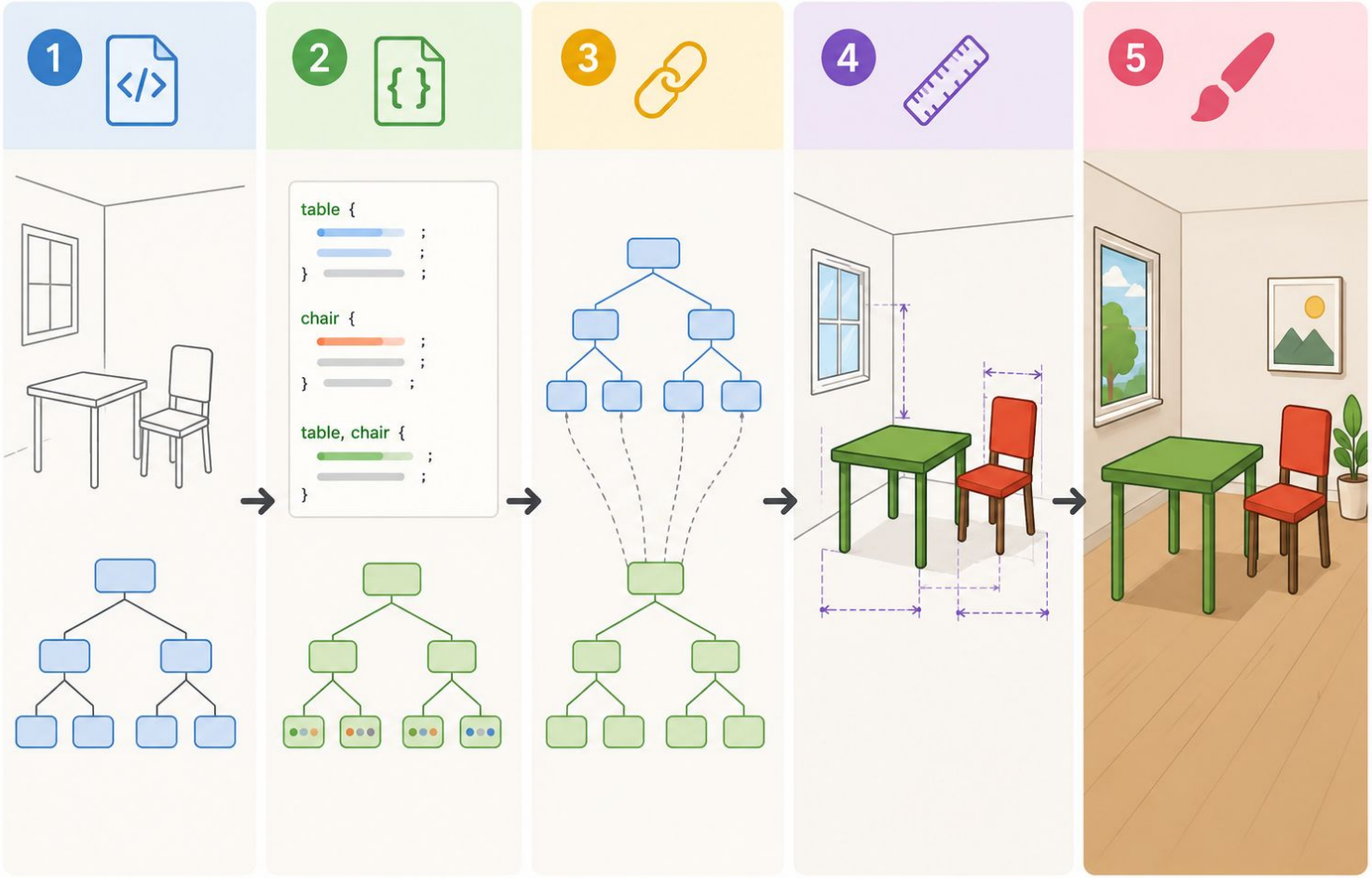


# Painting

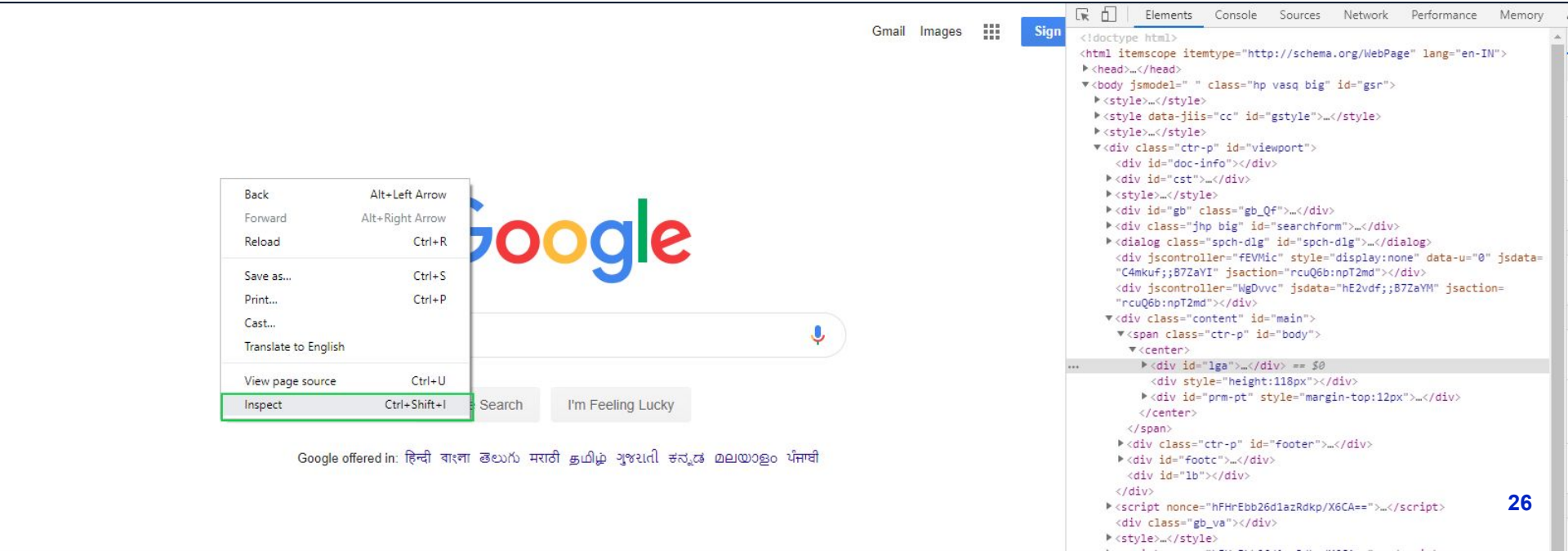
**Instructions on the order and location of elements on the page**

**Render the page**



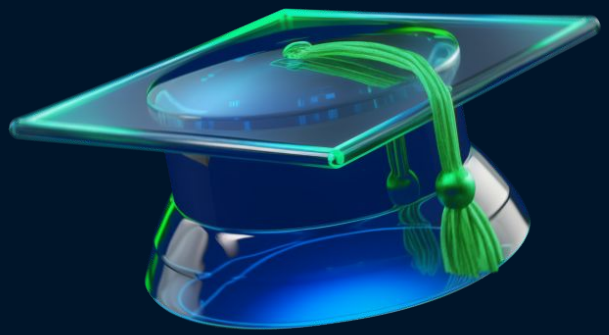

1
2
table {
}
chair {
}
table, chair {
}
3
4
5

# Chrome
# Dev Tools



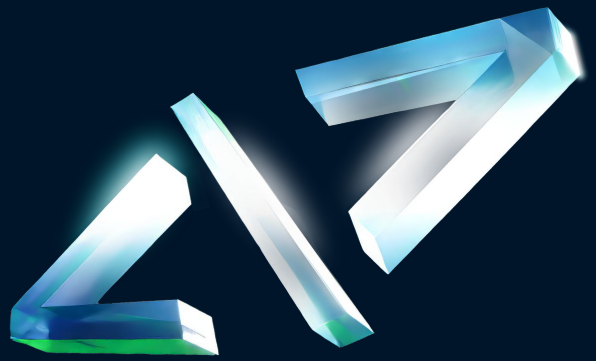
# What We Learned Today

1. What the Internet is
2. What happens when we want to load a website
3. Different types of browsers
4. How a browser renders a page



# Homework

1. Install required apps:
   - Figma
   - Slack
   - VS Code
2. Create accounts:
   - GitHub
   - Jira
   - CodePen
3. Read additional materials

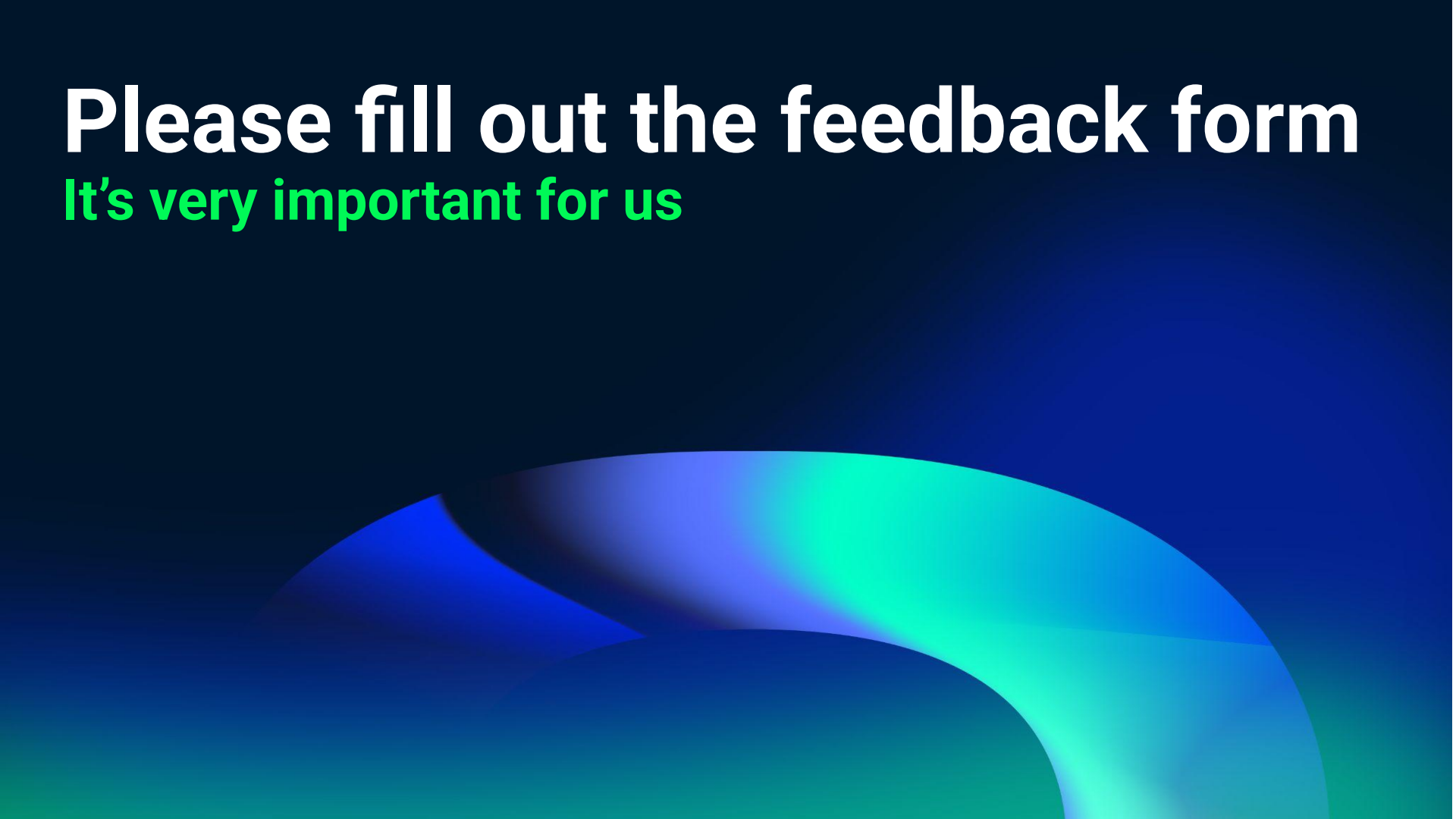


BRO Academy

# QUESTIONS?

# Please fill out the feedback form

## It’s very important for us



# THANK YOU!

# Have a good evening!